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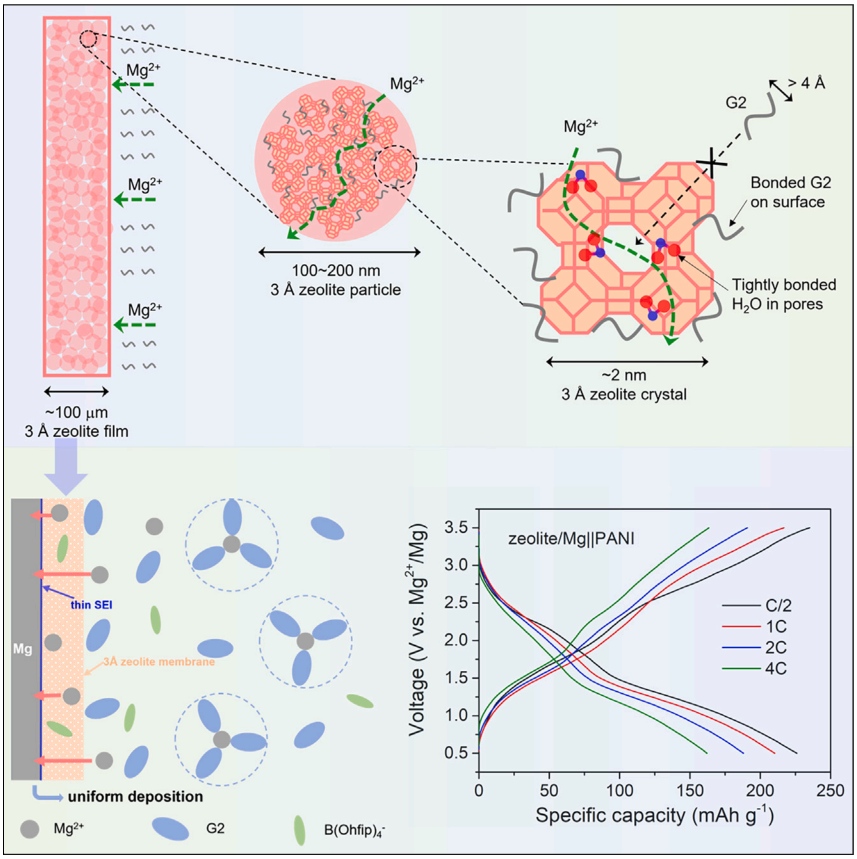

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Highlights

A zeolite 3A/polymer membrane functions as a protective interface for Mg anodes

Decomposition of free diglyme is suppressed at the Mg/electrolyte interface

Highly reversible Mg plating/stripping through the membrane over 6,000 h

Full cells show stable cycling at a 3.5 V cutoff voltage at high rates (4C)

Chloride-free Mg electrolytes are critical to achieve high-voltage Mg batteries without corrosion, but the short lifetime of Mg metal anodes induced by electrolyte decomposition presents a serious challenge. This work reports on a practical Mg^{2+}-ion conducting and cost-effective protective interface comprising a zeolite/polymer membrane that traps free solvent and thus limits its decomposition. Moreover, it assists the facile transport of Mg^{2+} cations, enabling fast, long-term plating/stripping of Mg through the interface, and supports full battery cycling at fast rates.

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Stabilizing magnesium plating by a low-cost inorganic surface membrane for high-voltage and high-power Mg batteries

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SUMMARY

Mg batteries with halide-free electrolytes suffer from poor stability of the Mg metal anode due to electrolyte decomposition. Here, we report a low-cost zeolite membrane supported on Mg to address this challenge. It vastly reduces the population (hence decomposition) of free diglyme at the Mg/electrolyte interface, while allowing facile transport of Mg^{2+} cations through the membrane. We demonstrate dendrite-free Mg plating/stripping performance in a magnesium tetrakis(hexafluoroisopropyloxy)borate/diglyme electrolyte with a 750-fold extended lifetime (over 6,000 h) and a high coulombic efficiency of ~98%. The prototype Mo_3S_4 cathode paired with the protected Mg anode shows 91% capacity retention over 200 cycles. Importantly, this membrane protects soluble species in a high-voltage organic polymer cathode from being reduced at the anode via shuttling, achieving a full cell with a 3.5 V cutoff voltage. This results in a high specific energy density of 320 Wh kg^{-1} and a power density of 1,320 W kg^{-1} based on cathode mass.

INTRODUCTION

Rechargeable Mg batteries (RMBs) are attractive as a "beyond lithium-ion battery" because of their use of a Mg metal anode.[1,2] Magnesium's advantages include a high theoretical capacity (3,883 mAh L^{-1} and 2,205 mAh g^{-1}), relatively low redox potential (−2.37 V vs. SHE) and high abundance (Mg is the 8th most abundant element and the 3rd most plentiful in seawater).[1,3,4] Mg was thought to deposit in a "dendrite-less" morphology in classical chloride-based electrolytes, unlike that commonly seen in other metal batteries (i.e., Li[5] and Zn[6]). However, this concept has been challenged by recent studies showing the existence of Mg dendrites during Mg plating[7] and uneven Mg stripping[8] in these halide electrolytes, where additional artificial interface design was reported to enable uniform Mg deposition.[9] Other concerns with halides arise from their low oxidative stability (usually <2.5 V vs. Mg^{2+}/Mg) and high corrosivity to battery components (i.e., current collectors and casings).[1,10]

To develop Mg electrolytes with high oxidative stability, a series of halide-free Mg salts have been studied recently, including magnesium bis(trifluoromethanesulfonimide) ($Mg(TFSI)_2$)[11–13] and newly invented alkoxy-borates/aluminates.[10,14–17] The electrolytes based on these salts in ethereal solvents can function up to 3.5 V.[1] Unfortunately, they exhibit pronounced decomposition during repeated Mg plating/stripping that leads to poor long-term reversibility.[11,12,18–23] Most of these Mg salts are electrochemical reductive during Mg plating and highly sensitive to the presence of moisture and impurities resulting from their synthesis procedures, which further

CONTEXT & SCALE

High-voltage Mg batteries with halide-free electrolytes suffer from poor stability of Mg metal anode due to electrolyte decomposition. We show here that this challenge can be addressed by creating a Mg^{2+}-conducting protective film on Mg metal, based on a low-cost zeolite-polymer membrane. Detailed spectroscopy studies in a $Mg(B(Ohfip)_4)_2$/diglyme electrolyte revealed that the population/decomposition of free diglyme molecules are greatly reduced at the Mg/electrolyte interface, whereas the membrane allows facile transport of Mg^{2+} cations that involves two chemically distinct migration pathways, both through the zeolite and on its surface. The protected Mg anode shows dendrite-free plating/stripping over long-term cycling (8 months). Furthermore, by simultaneously suppressing the shuttling of dissolved organic cathode monomers, this membrane enables Mg batteries to function at a high operation voltage of >3.5 V at a current rate of 4C and a power density of 1,320 W kg^{-1}.

aggravates their instability at the Mg anode.[18,24,25] In addition, the reductive instability of ether-based solvents has been widely reported as one of the major factors that induces electrolyte decomposition to polymeric components (for example, C–O or C=O species), leading to the formation of unstable solid-electrolyte interface (SEI) or passivation layer on Mg surface that induces inhomogeneous Mg plating/stripping with short cycle life.[11,12,19–21] The strategy of using lithium hexafluoroisopropoxy borate ($LiB(Ohfip)_4$) as an electrolyte additive has been reported to achieve long-term Mg plating/stripping in a $Mg(B(Ohfip)_4)_2$/1,2-dimethoxyethane (DME) electrolyte,[18] which is ascribed to the formation of a Li-species-containing SEI. However, the need for Li salts reduces the energy density and compromises the potential economic benefits anticipated for Mg batteries. Various artificial SEIs have also been reported to enable reversible Mg plating/stripping in $Mg(TFSI)_2$-based electrolytes by suppressing their decomposition and preventing the formation of a passivation film on the Mg surface[26–28]; however, a very low current density of less than 0.05 mA cm^{-2} was used in these studies. Chloride-salt induced protective interface on Mg were also widely reported to enable reversible and stable Mg^{2+}/Mg redox[29–32]; nonetheless, concerns remain on the dissolution of these chloride species from the Mg surface into electrolytes that may compromises the electrolyte oxidative stability. To the best of our knowledge, none of these studies reported full cell performances with an operating voltage of over 3 V. Therefore, creating a Mg^{2+}-conducting, practical and cost-effective protective interface from electrolyte decomposition remains an ongoing challenge for these high-voltage electrolytes.

Here, we utilize zeolite-polymer membranes to stabilize the Mg anode for the first time and achieve excellent performance in full cells at relatively high current rates. Zeolite A is used as molecular sieves for dehydration of gases or solvents owing to its propensity to absorb water.[33] Its cubic structure consists of aluminosilicate cages that form an 11.4 Å α-cavity and much smaller β cavity, which are interconnected in three dimensions to form the microporous structure. The windows into the cavities/pores range from 3–5 Å, as dictated by the cations that sit within them (K^+, Na^+, Ca^{2+}, etc.) to balance the negative charge of the aluminosilicate framework. For example, zeolite 3A—defined by a high K^+/Na^+ ratio—has windows of 3.2 Å in diameter (making it ideal for the adsorption of water), whereas zeolite 4A has windows of 3.8 Å, based on Na^+ as the only counter-cation. Zeolite 3A was recently reported to effectively create local concentrated electrolytes in the battery area.[34–37] For example, the oxidative activity of ethereal solvents in sodium-ion batteries was decreased based on a regulated electrolyte network[36]; incorporation of Na^+-diglyme (G2)-anion complexes in the zeolite pores; and suppressed activity of diglyme at the electrode/electrolyte interface. However, the functionality of the zeolite was not explicitly explained, while we note that diglyme is much too large to penetrate the 3.2 Å window (see below).

To establish proof of concept for our study, we chose a 0.25 M $Mg(B(Ohfip)_4)_2$/G2 electrolyte as a platform to study the solvent reduction activity during Mg plating/stripping, where the factor of salt decomposition can be alleviated. This is because bulky $B(Ohfip)_4^-$ anion results in less contact ion pairs compared with conventional $TFSI^-$ anion, which facilitates the fast interfacial charge transfer process with less anion reduction activity. $Mg(B(Ohfip)_4)_2$ is also a widely reported salt and can be synthesized with relatively lower cost and higher purity compared with other state-of-art salts (for example, $Mg(CB_{11}H_{12})_2$). In this study, we identify that the short lifetime of Mg plating/stripping in this electrolyte originates from the high activity of free G2 solvent molecules that reduce on the Mg/electrolyte interface to form a thick surface layer of decomposition by-products (Scheme 1). This induces inhomogeneous Mg

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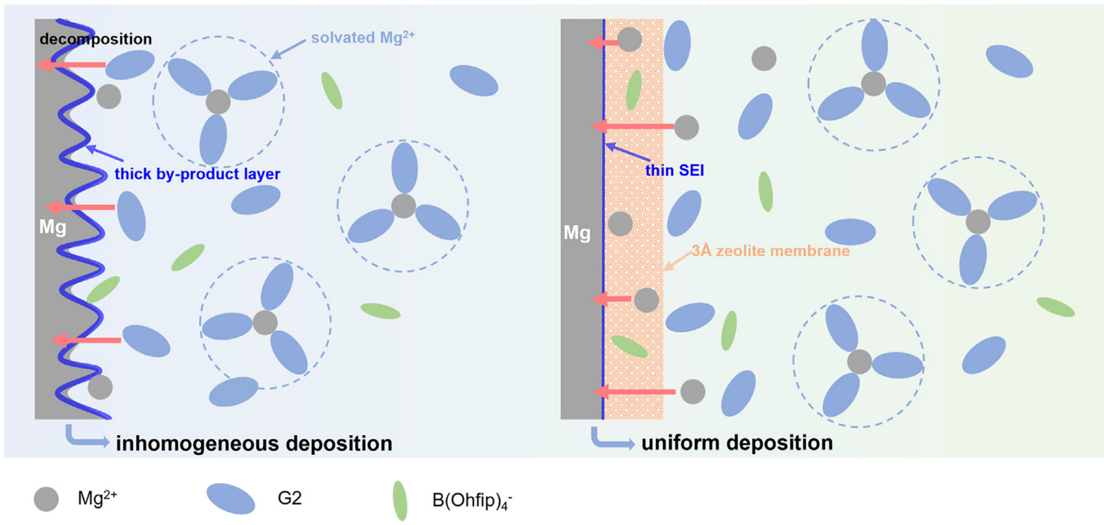


Scheme 1. Illustration of the functionality of the zeolite membrane
Inhomogeneous Mg deposition with pronounced solvent (G2) decomposition on a bare Mg/electrolyte interface (left), and uniform Mg deposition with suppressed G2 decomposition in the presence of 3A zeolite membrane on the Mg surface (right).

deposition and rapid cell failure. By using the low-cost zeolite film based on 3 Å molecular sieves as a protective interface on the Mg anode, G2 decomposition is effectively suppressed, generating a smooth Mg plating morphology and a thinner SEI. This is ascribed to the significantly alleviated free-G2 fraction within this interface revealed by Fourier transform infrared (FTIR) spectroscopy. The protected Mg anode thus achieves a 750-fold extended lifetime in Mg||Mg symmetric cells (over 6,000 h/8.3 months) at a current density of 0.3 mA cm^{-2} and an areal capacity of 0.3 mAh cm^{-2}. Two chemically distinct Mg^{2+} environments were revealed by ^{25}Mg magic angle spinning nuclear magnetic resonance (MAS NMR) spectroscopy, which are likely involved in the Mg^{2+} migration pathways through the zeolite membrane. A prototype full cell based on a Mo_3S_4 cathode and ethereal electrolyte exhibits excellent capacity retention of 91% over 200 cycles. We furthermore demonstrate Mg batteries that are very stable at a high operation voltage of >3.5 V vs. Mg^{2+}/Mg at a 2C rate using polyaniline (PANI) as the cathode. By simultaneously suppressing the shuttling of dissolved PANI monomers with the zeolite membrane, a high energy density up to 320 Wh kg_{PANI}^{-1} and a power density of 1,320 W kg_{PANI}^{-1} are achieved.

RESULTS AND DISCUSSION

Zeolite 3A membrane to suppress G2 activity

The fabricated flexible zeolite membrane shows a flat surface produced by ball-milled 3A molecular sieve particles that have a dimension less than 200 nm (Figures 1A and S1). Different from previous reported defect-free metal-organic-framework (MOF)-based interfacial membrane that completely blocks the electrolyte penetration,[26] the existence of many macropores between the zeolite particles enables good absorption of the electrolyte, thus ensuring fast ion transport with a high ionic conductivity of 0.76 mS cm^{-1} (Figure S2). After soaking this zeolite membrane in 0.25 M $Mg(B(Ohfip)_4)_2$)/G2 electrolyte (denoted as zeolite/0.25 M), the change in bonding environment of the G2 molecules was revealed by FTIR spectroscopy. A gradual blue shift of the CH_2 bending mode (G2) with increased salt concentration can be observed (Figure S3, from 852 cm^{-1} in pure G2 to 858 cm^{-1} in 0.25 M $Mg(B(Ohfip)_4)_2$/G2). Addition of the zeolite membrane (zeolite/0.25 M) leads to a

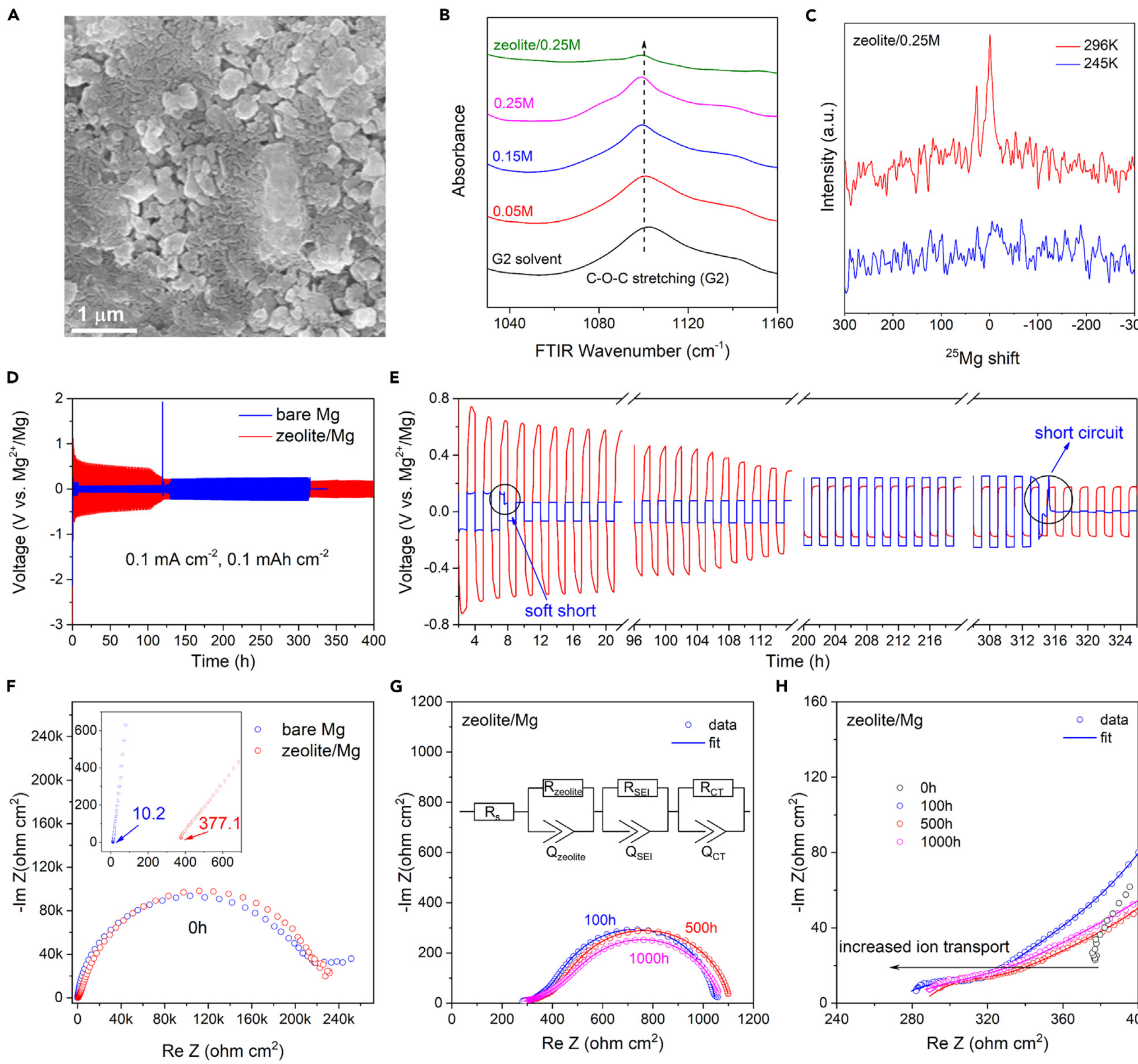


Figure 1. Characterization of the zeolite membrane

(A) SEM image of the zeolite membrane.

(B) FTIR spectra at different concentrations of $Mg(B(Ohfip)_4)_2$/G2 electrolyte and zeolite-modified 0.25 M electrolyte.

(C) Variable temperature solid-state ^{25}Mg MAS NMR spectra of zeolite/0.25 M.

(D and E) Galvanostatic Mg plating/stripping of Mg||Mg symmetric cells with bare Mg and zeolite/Mg at 0.1 mA cm^{-2} (0.1 mAh cm^{-2}) and (E) magnified voltage curves.

(F) Nyquist plots of symmetric cells at open circuit with zeolite/Mg and bare Mg.

(G and H) Nyquist plots of symmetric cells with zeolite/Mg after plating/stripping for different hours at 0.3 mA cm^{-2} (0.3 mAh cm^{-2}): the evolution of (G) interfacial resistance fitted (shown by line, see Table S1) with the indicated equivalent circuit and (H) bulk ion transport.

further blue shift to 861 cm^{-1}, indicating that the free G2 molecules in the electrolyte are adsorbed by the zeolite membrane. This is further verified by the change of the C-O-C stretching mode at ~1,101 cm^{-1} (Figure 1B). The gentle decrease in its intensity as the salt concentration increases originates from G2 molecules being bound to Mg^{2+} ions.[38] The pronounced decrease of this peak in zeolite/0.25 M, however, suggests the existence of much more tightly bound (i.e., less free) G2 molecules.

Therefore, we can anticipate lower decomposition of free G2 at the Mg/zeolite interface during Mg plating/stripping.

We carried out solid-state ^{25}Mg MAS NMR on a zeolite/PTFE film to gain insight into the Mg^{2+} environments in zeolite/0.25 M (Figure 1C). As the film was dried at 120°C prior to analysis, we expect the presence of indigenous water in the zeolite pores, which is not removed until >300°C. At 296 K under MAS conditions, two ^{25}Mg peaks were detected at 0.2 and 27.2 ppm. These peaks were, respectively, assigned to Mg^{2+} coordinated to water molecules—hydrated Mg^{2+} in the zeolite structure—and Mg^{2+} coordinated to diglyme molecules (presumably Mg-O_6 octahedral units). The former assignment is close and consistent with free Mg^{2+} in 5 M $MgCl_2$ aqueous solution (see experimental procedures), and the latter is in excellent agreement with the shift for Mg^{2+}-diglyme coordination in a xerogel (24.7 ppm).[39] These environments are represented, according to the peak integrals, by a molar ratio of 1:3 in zeolite/0.25 M. At 245 K with identical parameters including the number of scans, however, no ^{25}Mg peaks, but only noise was observed in the spectrum. We assume that the motional linewidth narrowing is significant at 296 K, whereas the linewidth broadening due to lack of motion for both hydrated and diglyme-coordinated Mg^{2+} at 245 K renders the broad peaks buried in the noise of the ^{25}Mg spectrum. Therefore, it is reasonable to propose significant mobility for both Mg^{2+}-coordination environments in zeolite/0.25 M at room temperature at the local level, possibly giving rise to two different Mg^{2+} migration pathways (see discussion below).

The effectiveness of the zeolite membrane in stabilizing Mg plating/stripping was tested in Mg||Mg symmetric cells using 0.25 M $Mg(B(Ohfip)_4)_2$/G2 as the electrolyte. Figure 1D displays the initial 400-h Mg plating/stripping behavior using bare Mg and zeolite-protected Mg (zeolite/Mg) at a typical current density of 0.1 mA cm^{-2} with an areal capacity of 0.1 mAh cm^{-2}. In the presence of zeolite, an initial activation process is observed with a Mg plating overpotential of 0.74 V (Figure 1E). The overpotential decreases to a stable value of 0.17 V after the first 120-h activation, which is only slightly higher than that of initially using bare Mg (~0.13 V). The slightly higher plating overpotential can be ascribed to the insulating nature of the zeolite membrane and lower ion transport (see below), which increase the interfacial resistance during Mg plating/stripping. However, with bare Mg, a sudden voltage drop to 0.06 V occurs after only 7 h of plating/stripping, which is a clear indication of the occurrence of soft shorts. This results in the fluctuation of the Mg overpotential in subsequent cycling until a complete short circuit is observed after 314 h. In contrast, cells with zeolite/Mg show stable Mg plating/stripping without any abrupt voltage change, exhibiting excellent stability.

We conducted electrochemical impedance spectroscopy (EIS) studies to identify interfacial stability. Figure 1F reveals that both cells with bare Mg and zeolite/Mg show an initial large interfacial resistance that can be ascribed to the native MgO.[40] With the presence of zeolite, a much higher resistance (377.1 ohm cm^2) related to ion transport is observed compared with only 10.2 ohm cm^2 for bare Mg, which explains the initial higher overpotential for Mg plating/stripping in Figure 1D. In subsequent plating/stripping processes, the interfacial resistance for a bare Mg anode quickly decreases to near zero after only 10 h (Figure S4), again, a clear indication of short circuits that is consistent with the sudden decrease in plating/stripping overpotential (Figures 1D and 1E). With zeolite/Mg anode, however, both interfacial resistance (Figure 1G) and ion transport resistance (Figure 1H) decrease during the electrochemical conditioning process that typically exists in Mg electrolytes.[41] Compared with other reported interfacial strategies to stabilize Mg plating/stripping (Table S1), zeolite/Mg in 0.25 M $Mg(B(Ohfip)_4)_2$)/G2 shows much lower interfacial resistance. The gradually increased bulk ion conductivity is probably due to the electrolyte wetting and/or reconfiguration of the Mg^{2+} solvation structure within the zeolite membrane. To further investigate the electrochemical process at electrolyte-zeolite/Mg interface, we fitted the impedance with a bulk electrolyte resistance (R_s) and three interfacial resistances (Figure 1G): zeolite membrane resistance ($R_{zeolite}$) at high frequency, SEI resistance (R_{SEI}) at medium frequency, and charge transfer resistance (R_{CT}) at low frequency. These resistances remain stable at different hours (Table S1), confirming the superiority of zeolite membrane to stabilize Mg plating/stripping during long-term cycling.

Highly reversible Mg plating/stripping

The Mg plating/stripping behavior was first evaluated by cyclic voltammetry (CV) on a Cu electrode (Figure S5). For zeolite/Mg anode, the onsets of cathodic/anodic peaks are similar to that of bare Mg, suggesting that the presence of this insulating membrane has minor impact to the kinetics of Mg^{2+}/Mg redox. To further examine the reversibility of the zeolite/Mg anode under realistic conditions, we galvanostatically measured the coulombic efficiency (CE) of Mg plating/stripping of asymmetric cells on a Cu substrate at a moderate current density of 0.3 mA cm^{-2} and an areal capacity of 0.3 mAh cm^{-2} (Figure 2). The CE for zeolite/Mg quickly increases to >98% after the initial 10 cycles and is sustained for over 160 cycles (Figures 2A and 2B). With bare Mg, the Mg||Cu cell fails after only 39 cycles due to short circuits (Figures 2A and S6). The CE was also measured with a modified Aurbach method in asymmetric cells,[42] with an areal capacity of 1 mAh cm^{-2} as a Mg reservoir. With zeolite/Mg, an average 96.3% CE can be obtained (Figure 2C), in contrast to the quick shorts with bare Mg (Figure S7). The long-term stability of the zeolite/Mg anode was further evaluated in symmetric cells. We record a cell cycling lifetime of over 6,000 h (8.3 months or 3,000 cycles) without significant increase of polarization: more than 750-fold vs. that of bare Mg (Figure 2D). We note the plating/stripping curves with zeolite membrane gradually changed to semi-rectangular shape during the activation process and remain constant for thousands of hours, where the possibility of soft short has been ruled out by previous impedance studies (Figures 1F–1H). The origin of this change is probably due to the difference in nucleation/growth model between initial Mg plating/stripping and after activation, which is beyond the scope of this study and will be examined in future studies. To the best of our knowledge, we believe this is one of the longest lifetimes of a Mg anode in organoborate or chloride-based electrolyte (Table S2). At more challenging conditions of 0.5 mA cm^{-2} and 0.5 mAh cm^{-2}, the cell still shows very stable cycling for over 2,500 h (3.5 months or 1,250 cycles; Figure S8).

Dendrite-free Mg plating morphology

To examine the Mg deposition morphology under the protection of zeolite, Cu electrodes with plated Mg were carefully retrieved and analyzed by scanning electron microscopy (SEM), where insulating zeolite membrane was peeled off from Cu collector prior to the imaging experiments. A smooth and homogeneous Mg deposit was observed on Cu protected by zeolite membrane (Figures 3A–3C), showing a two-dimensional (2D) Mg deposition layer merged by micro-size globules (Figure 3B) with nanoscale Mg nuclei (Figure 3A). For the Mg deposited on the bare Cu collector (Figures 3D–3F), to the contrary, an inhomogeneous and three-dimensional (3D) island (Figure 3E) was observed with much larger Mg nuclei (Figure 3D). Such a 3D island growth of Mg was reported with similar negative impact to dendrite growth and can easily induce the early-onset shorting of asymmetric cells (Figure 2A).[13] The

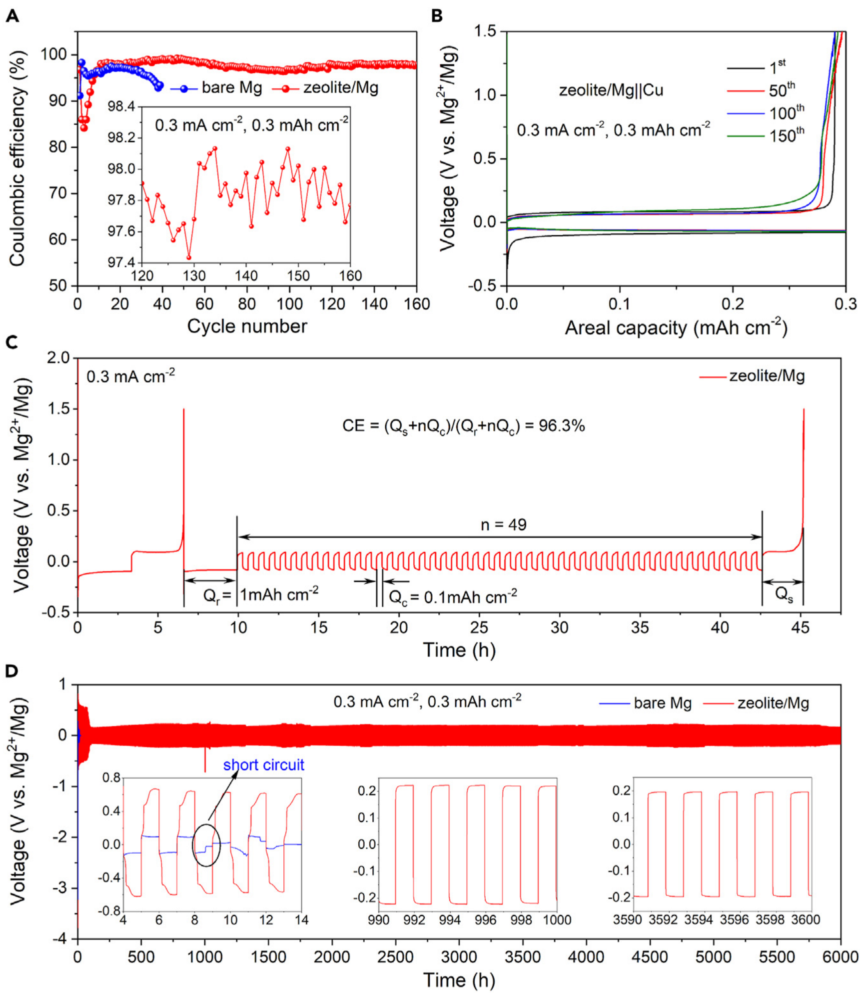


Figure 2. Electrochemical studies of Mg plating/stripping

(A and B) Evolution of coulombic efficiency on cycling in Cu||Mg and Cu||zeolite/Mg asymmetric cells and (B) corresponding voltage profiles of Cu||zeolite/Mg. The zeolite/Mg was activated in a symmetric cell for 20 cycles prior to being coupled with Cu.

(C) Modified Aurbach measurement[42] of coulombic efficiency in a Cu||zeolite/Mg cell.

(D) Long-term cycling of Mg||Mg and zeolite/Mg||zeolite/Mg symmetric cells. Cycling conditions are at 0.3 mA cm^{-2} with a stripping/plating capacity of 0.3 mAh cm^{-2}.

undesirable morphology was also observed on cycled bare Mg metal foil (Figure S9) that can be partly ascribed to the high reactivity of free G2 solvent molecules, which decompose on the Mg surface (see X-ray photoelectron spectroscopy [XPS] studies below). In contrast, by limiting the free G2 activity using a 3 Å zeolite membrane, the solvent decomposition reaction is suppressed and yields a very smooth Mg deposition morphology (Figures 3A–3C and S9). It should be noted that the 3 Å pore size of the selective molecular sieves plays a key role in enabling stable cycling of these cells, as the use of 4 Å zeolite membrane leads to very poor Mg plating/stripping performance and undesirable deposition morphologies (Figure S10). This is discussed in detail below.

XPS studies of interfacial SEIs

The surface composition of the deposited Mg is directly correlated to electrolyte decomposition. To prove this, we carried out high-resolution XPS on the topmost

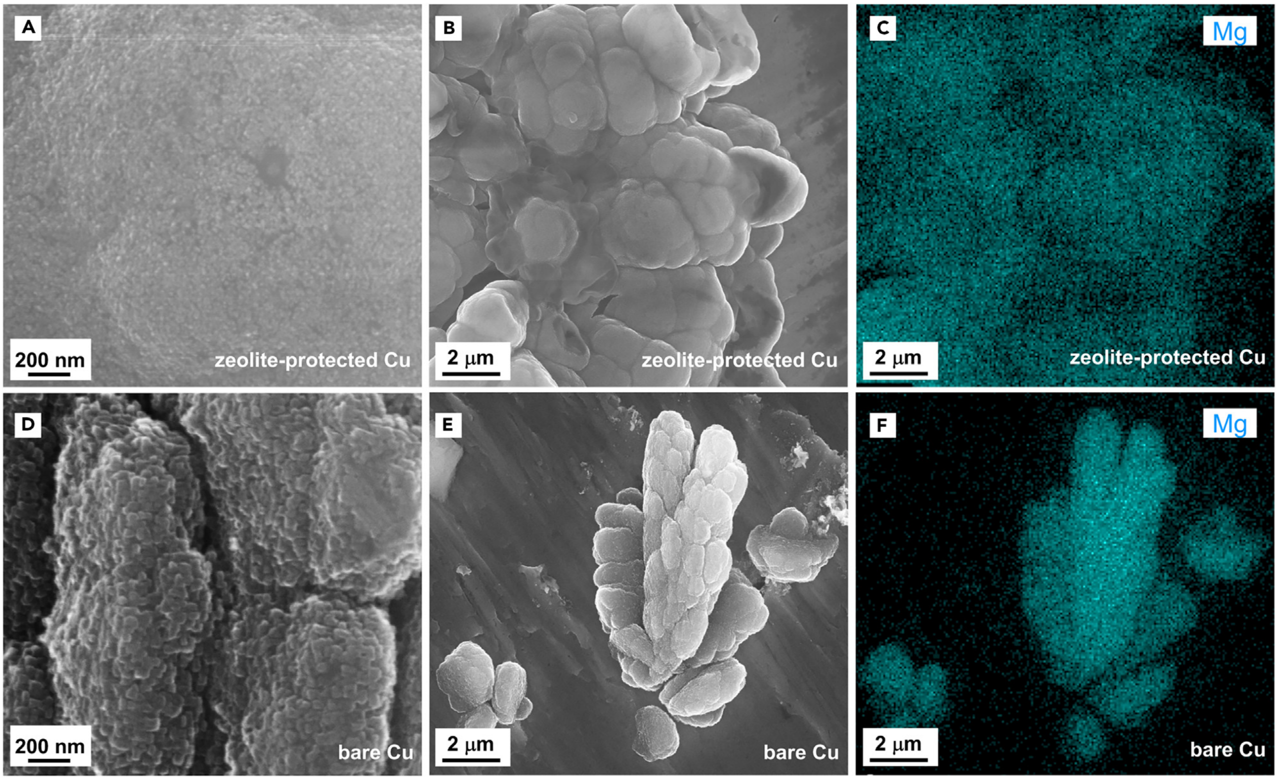


Figure 3. Mg plating morphologies on Cu electrodes

(A, B, D, and E) SEM images and (C and D) corresponding EDX maps of Cu electrodes with deposited Mg at a current density of 0.3 mA cm^{-2} (0.3 mAh cm^{-2}): (A–C) zeolite-protected Cu and (D–F) bare Cu. Please note the zeolite membrane was removed prior to image the Mg morphology on zeolite-protected Cu.

and Ar^+-sputtered surface of retrieved Mg electrodes with/without zeolite protection. As shown in Figure 4A, in the O *1s* region of bare Mg, a peak at 532.7 eV corresponding to polymeric/organic C=O/C–O accounts for over 62% of the peak area, along with features in the C *1s* region (C–O at 286.1 eV [51%] and C=O at 289.5 eV [11 %]). These can be ascribed to the decomposition of G2 at the Mg/electrolyte interface.[12,20] Two additional peaks for $Mg(OH)_2$ (531.7 eV, 32%) and MgO (530.2 eV, 6%) were also identified, which originate from the native oxide passivation surface or trace moisture in the electrolyte. MgF_2 (685.9 eV) and B-O (192.5 eV) species that result from $B(Ohfip)_4^-$ decomposition were also confirmed in the F *1s* and B *1s* region (Figure S11). Whereas for zeolite/Mg, the intensity of C–O/C=O peaks related to G2 decomposition significantly decreases in both the C *1s* and O *1s* region (Figure 4B), indicating alleviated side reactions. After sputtering for 120 s, the percentage of these species for both bare and protected Mg significantly decreased (Figure 4C); however, the polymeric/organic fraction for zeolite/Mg is still much less than that of bare Mg. This is further evidence of the effectiveness of the zeolite membrane in protecting Mg from G2 decomposition on plating/stripping.

The proposed mechanism of suppressed G2 decomposition and Mg^{2+} diffusion pathway via the zeolite-3A membrane is shown in Scheme 2. The G2 molecule in the electrolyte cannot enter the zeolite through its 3 Å windows due to its large molecular size (>4 Å diameter of the linear chain), as verified by thermogravimetric analysis (TGA, Figure S12). However, glyme has been shown to strongly adsorb on the surface of zeolite crystals by interaction between the methoxy groups of G2 and

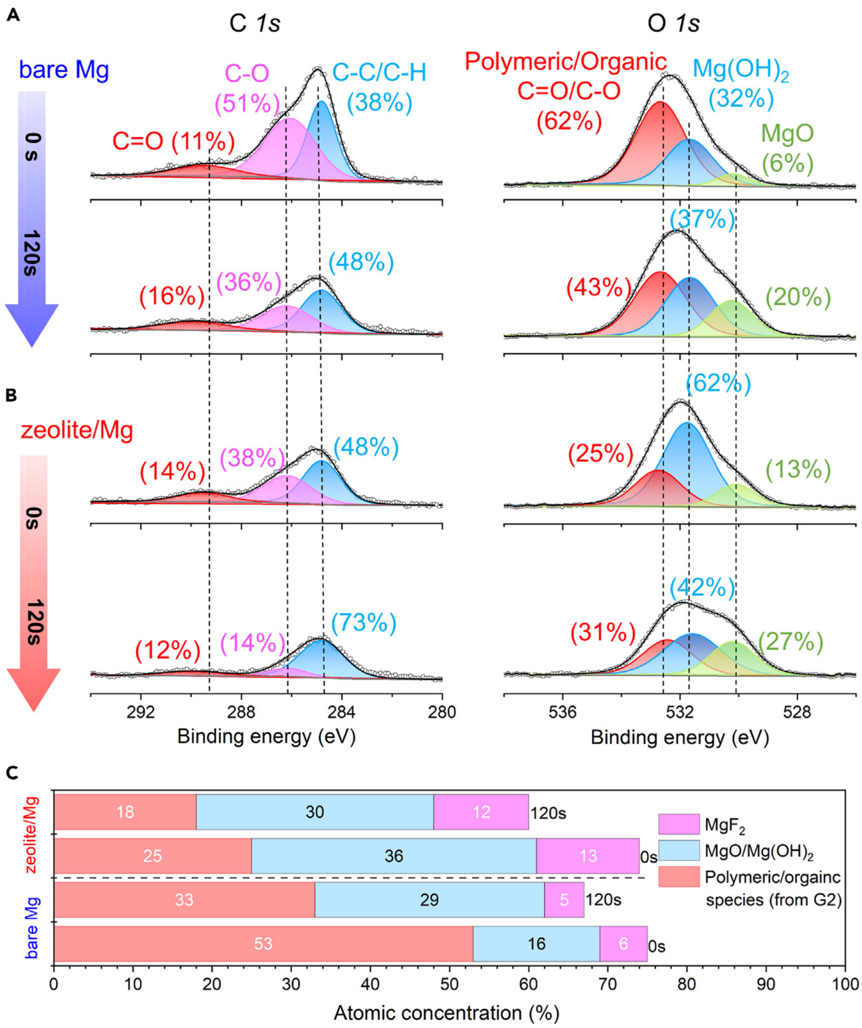
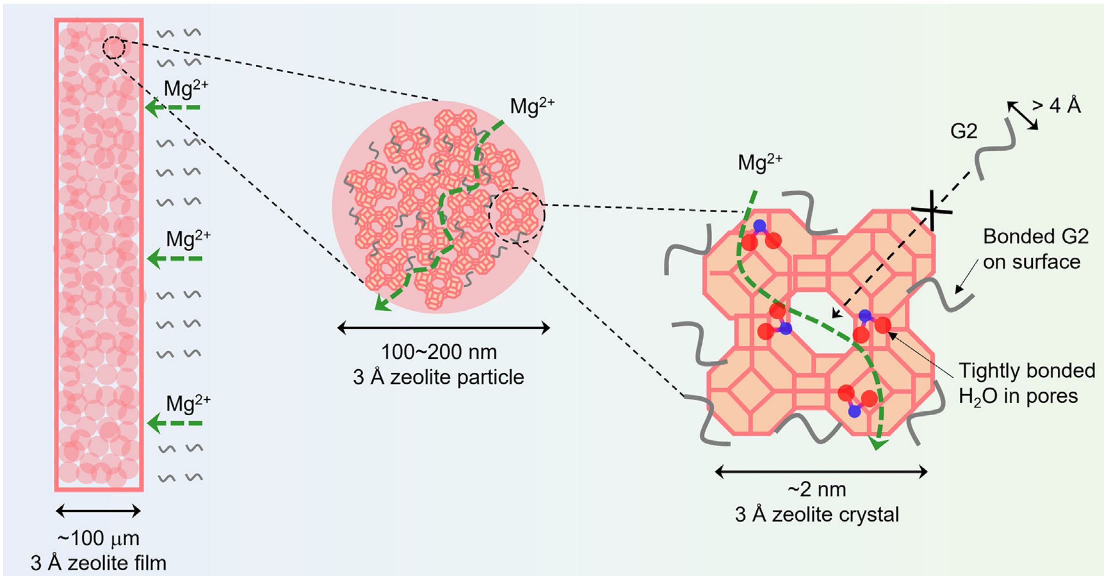


Figure 4. Interfacial XPS studies

(A and B) Depth-profiling XPS spectra in the C *1s* and O *1s* regions for Mg plated on the 10th cycle on (A) bare and (B) zeolite-protected anodes at 0.3 mA cm^{-2} (0.3 mAh cm^{-2}).

(C) The relative fractions of different species identified on the cycled electrodes. The peaks of C=O and C–O in the C1s and O1s spectra correspond to polymeric/organic decomposition products from the G2 solvent.[11,12,20,21]

the oxygen atoms of the zeolite.[43] Indeed, the existence of little free G2 content in the zeolite 3A membrane was clearly shown by FTIR (Figure 1B). Such surface binding favorably reduces the decomposition of G2 available at the interface (Figure 4). Furthermore, strong interaction of G2 with the zeolite surface would promote desolvation of the diglyme coordinated to Mg^{2+}. In turn, this could promote diffusion of Mg^{2+} along the zeolite crystal surface. In addition, significant fractions of Mg^{2+} are readily exchanged with K^+ within the zeolite at room temperature (Table S3). This suggests the existence of a possible second Mg^{2+} diffusion pathway within the zeolite crystal,[44] which could take place by hopping along H_2O molecules in the cages.[35] Two different Mg^{2+} diffusion pathways were consistent with the solid-state ^{25}Mg MAS NMR results (Figure 1C), with both G2-solvated and hydrated Mg^{2+} showing significant mobility at room temperature. We note that those tightly bonded H_2O molecules in zeolite 3A can only be removed well above 300°C



Scheme 2. Proposed mechanism of suppressed G2 activity and Mg^{2+} diffusion pathway

The decreased free G2 in the zeolite membrane is due to surface adsorption by methoxy groups of G2 on zeolite. Mg^{2+} can diffuse either through surface adsorbed G2 along zeolite surface, or through the pores in the zeolite crystals with the assistance of transport via H_2O molecules, which reside within the cages.

(Figure S13) and thus are not free to react with the Mg surface. Indeed, using zeolite 4A (where the water in the α cages is more loosely bound, see Figure S14) in place of 3A reproducibly results in almost immediate cell failure (Figure S10). We speculate that here, the H_2O exchanges into the electrolyte to passivate the Mg anode by forming $Mg(OH)_2$/MgO.

Electrochemistry of Mg||Mo_3S_4 and Mg||PANI full cells

The stability of the zeolite/Mg anode was further evaluated in prototype full cells with Chevrel-phase Mo_3S_4 as the cathode (Figure S15), a widely reported material for nonaqueous Mg batteries.[3,45,46] We first measured the CV curves of the Mo_3S_4||Mg and Mg_3S_4||zeolite/Mg cells at a scan rate of 0.1 mV s^{-1} (Figure 5A). Both cells display similar redox behavior characteristic of the Chevrel phase; however, a slightly higher polarization is observed for zeolite/Mg due to the insulating nature of the zeolite membrane. The Mo_3S_4||zeolite/Mg cell was investigated under galvanostatic control at rates of C/20 (1C = 128 mAh $g_{Mo3S4}{}^{-1}$), C/10 and C/5, providing discharge capacities of 96, 88, and 76 mAh $g_{Mo3S4}{}^{-1}$, respectively (Figure 5B). This rate performance is similar to that of a Mo_3S_4||Mg cell (Figure S16), showing good compatibility between the zeolite membrane and Mg anode under realistic full cell conditions. The long-term cycling stability of these full cells was evaluated at C/5 in the voltage range of 0.2–2.4 V under the same conditions. As shown in Figure 5C, Data S1, and Data S2, in contrast to the rapid failure of Mg||Mo_3S_4 after only 16 cycles (Figure S17), the zeolite/Mg||Mo_3S_4 cell shows stable cycling for over 200 cycles with a capacity retention of 91% (Figure S18). It should be noted that Mo_3S_4 functions very well in $MgCl_2$-based electrolytes,[20,45] where the Mg anode can sustain long-term Mg plating/stripping. The observed rapid shorting of the Mo_3S_4||Mg cell in this study is mainly due to the poor stability of the bare Mg anode in $Mg(B(Ohfip)_4)_2$/G2.

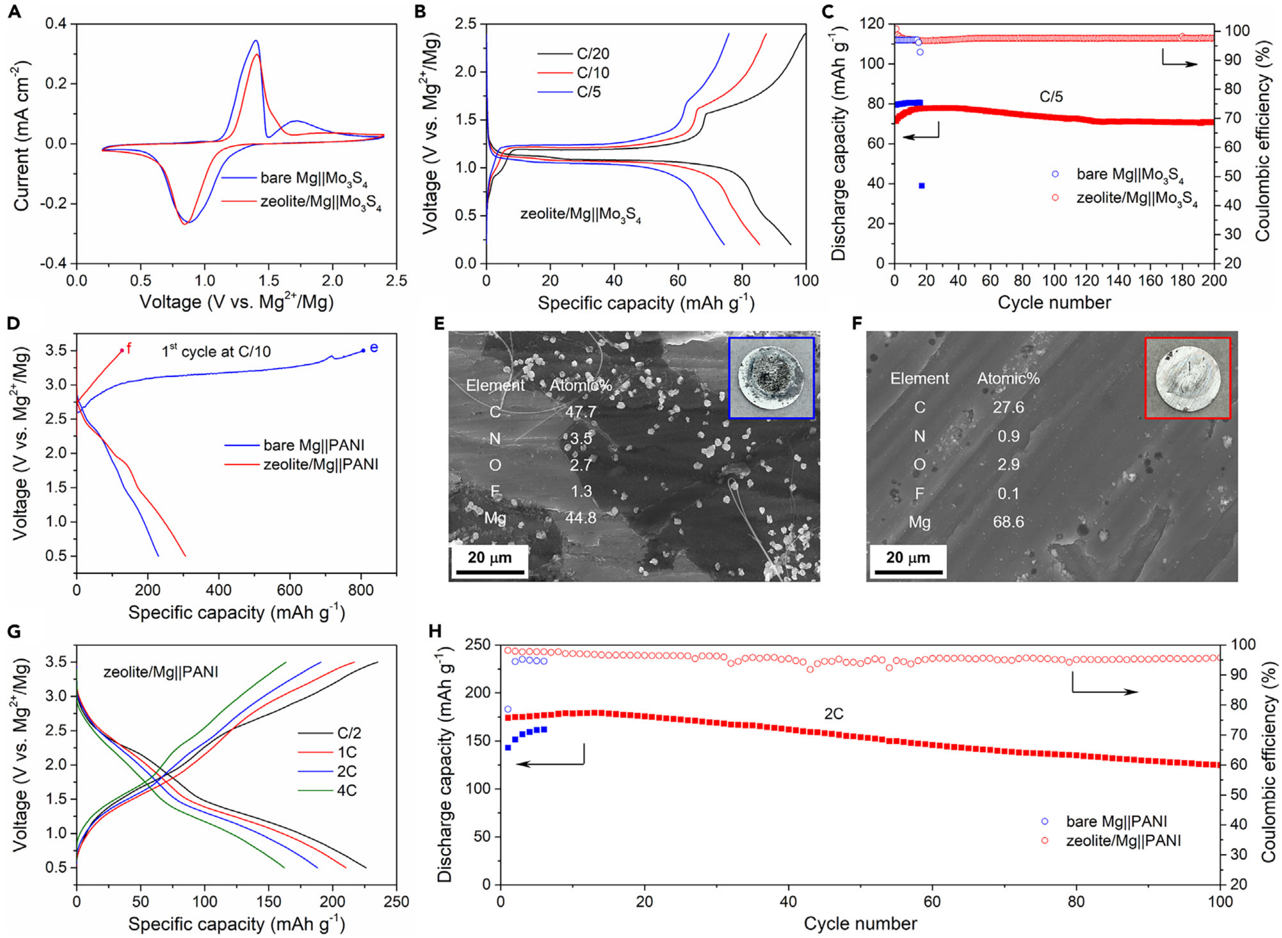


Figure 5. Electrochemical performance of full cells

(A) CV curves of Mo_3S_4 cathodes with bare Mg and zeolite/Mg at a scan rate of 0.1 mV s^{-1}.

(B) Voltage profiles of Mo_3S_4|| zeolite/Mg at different rates (1C = 128 mAh g_{Mo3S4}^{-1}).

(C) Discharge capacity and corresponding coulombic efficiency of a Mo_3S_4 cathode with bare Mg and zeolite/Mg as a function of cycle number at C/5.

(D) First cycle of PANI||Mg and PANI|| zeolite/Mg at C/10 (1C = 250 mA g_{PANI}^{-1}).

(E and F) SEM images and element ratios measured by EDX of (E) bare Mg and (F) zeolite-protected Mg that were retrieved from full cells after the initial charge.

(G) Voltage profiles of PANI||zeolite/Mg at different rates.

(H) Discharge capacity and corresponding coulombic efficiency of PANI||zeolite/Mg and PANI||Mg as a function of cycle number at a 2C rate. Prior to rate and long-term cycling measurements, all Mo_3S_4 and PANI cathodes were activated at C/20 and C/10, respectively.

To evaluate the stability of the zeolite/Mg anode in high-voltage cells, we employed organic polyaniline (PANI) as a platform cathode and measured the electrochemical performance over a voltage range of 0.5–3.5 V. Cells with zeolite/Mg||PANI exhibited a charge capacity of 127 mAh g_{PANI}^{-1} at C/10 (1C = 250 mA g_{PANI}^{-1}; Figure 5D). By reducing fully oxidized PANI to its reduced state at 0.5 V in the subsequent discharge process, a much higher discharge capacity of 305 mAh g_{PANI}^{-1} was obtained due to the association of Mg^{2+} cations with PANI (Figure S19). In contrast, an unprotected Mg anode PANI||Mg cell shows a very high initial charge capacity of over 806 mAh g_{PANI}^{-1}, with a subsequent discharge capacity of only 230 mAh g_{PANI}^{-1}. The observed overcharge does not originate from electrolyte oxidation (Figure S20), but from the aggravated dissolution/reduction of residual PANI monomers in the electrolyte (Figure S21), which shuttle through the glassfiber separator to the anode side and react with Mg to form a carbon/nitrogen-rich interface (Figure 5E). Such consumption of monomers on the anode side thus leads to the decreased discharge capacity in the subsequent process. In the presence of the zeolite membrane on the Mg surface (Figure 5F), the shuttling effect is significantly suppressed, as verified by the much lower fraction of carbon/nitrogen species in the EDX analysis and cleaner Mg surface after the first charge. After initial activation at C/10, the zeolite/Mg||PANI cell shows a high discharge capacity of 226 mAh g_{PANI}^{-1} at C/2 with an average discharge voltage of 1.4 V (Figure 5G). This correlates to a high energy density up to ~320 Wh kg_{PANI}^{-1}. In addition, we note that Mg^{2+} cation association/dissociation chemistry dominates the energy storage process (Figure S19), which is different from previous reported anion storage chemistry in chloride-based[47] and $Mg(TFSI)_2$-based electrolyte.[48] This likely originates from the bulky $B(Ohfip)_4)_2^-$ anion that only weakly associates with PANI, compared with the smaller and electron localized Cl^-/$TFSI^-$ anion. Nonetheless, the cation-type association/dissociation chemistry of organic PANI still obviates the sluggish Mg^{2+} solid-state diffusion exhibited in intercalative-type inorganic cathodes.[49,50] This results in a very good rate capability with a specific capacity of 210, 188, and 162 mAh g_{PANI}^{-1} at 1C, 2C, and 4C, respectively, and a power density of 1,320 W kg_{PANI}^{-1} at a 4C rate. Figures 5H and S22, Data S3, and Data S4 show the long-term cycling of a PANI||zeolite/Mg full cell at a moderate rate of 2C, demonstrating stable cycling with ~70% capacity retention for over 100 cycles, in contrast to the quick failure of PANI|| Mg due to both aggressive dissolution/shuttling of PANI and instability of bare Mg (Figure S23).

Conclusions

We report a low-cost zeolite membrane as an effective protective interface for a Mg metal anode, leading to dendrite-free and highly stable Mg plating/stripping in a 0.25 M $Mg(B(Ohfip)_4)_2$/G2 electrolyte with a ultralong lifetime of over 6,000 h (>8 months). The function of this zeolite membrane is to reduce the free solvent content on the Mg/electrolyte interface, resulting in alleviated solvent decomposition. The superiority of the zeolite membrane-protected Mg anode is verified in full cells with not only a prototype Mo_3S_4 cathode but also a high-voltage PANI cathode owing to suppressed shuttling of free monomers. This strategy represents a viable and cost-effective avenue for stabilizing plating/stripping of metal anodes via interface modification, thus paving the way for the development of high-voltage magnesium-metal batteries.

EXPERIMENTAL PROCEDURES

Resource availability

Lead contact

Further information and requests for resources and materials should be directed to and will be fulfilled by the lead contact, Linda F. Nazar (lfnazar@uwaterloo.ca).

Materials availability

This study did not generate new unique materials.

Data and code availability

The datasets generated in this study are available from the lead contact on reasonable request.

Synthesis of magnesium hexafluoroisopropoxide ($Mg(Ohfip)_2$)

This material was synthesized using a similar procedure to that previously reported,[51] but magnesium ethoxide (98% Sigma-Aldrich) was used instead of calcium methoxide. ^{1}H NMR (300 MHz, CD3CN): δ = 4.85(m). ^{19}F NMR (282 MHz CD3CN): δ = −76.8.

Synthesis tris-hexafluoroisoproxy borate ($B(Ohfip)_3$)

Tris-hexafluoroisopropoxy borate was prepared according to a previously reported procedure,[52] and chemical analysis was in agreement with the expected composition. This borate salt was sublimed before use.

Synthesis of magnesium hexafluoroisoproxy borate ($Mg(B(Ohfip)_4)_2 \cdot 3DME$)

1,2-Dimethoxy ethane (DME, 99% Sigma-Aldrich) was first passed over a column of activated alumina, distilled over sodium, degassed, and stored under molecular sieves. Magnesium hexafluoroisopropoxide ($Mg(Ohfip)_2$) and tris-hexafluoroisopropoxy borate ($B(Ohfip)_3$) were mixed in the appropriate ratio in DME at 65°C for 6 h. The clear solution obtained was then concentrated under dynamic vacuum at room temperature to obtain the $Mg(B(Ohfip)_4)_2 \cdot xDME$ salt. The salt was then redissolved in DME and recrystallized with hexanes (mixture of isomers 95% Sigma-Aldrich) to obtain crystals of $Mg(B(Ohfip)_4)_2 \cdot 3DME$. ^{1}H NMR (300 MHz, CD3CN): δ = 4.76(m), 3.55(t), 3.42(s). ^{19}F NMR (282 MHz CD3CN): δ = −75.1 11B NMR (96.25 MHz, CD3CN): δ = 9.2(s).

Preparation of 0.25 M $Mg(B(Ohfip)_4)_2$ electrolyte solution

The weighted $Mg(B(Ohfip)_4)_2 \cdot 3DME$ powder was dissolved into proper amount of G2 and was stirred overnight at 25°C. The obtained clear electrolyte solution was directly used for cell assembly without any pre-conditioning process.

Preparation of zeolite membrane

Three grams of 3 Å molecular sieves (Sigma-Aldrich, 8–12 mesh) were ball-milled at 400 rpm for 12 h before being dried in vacuum at 200°C overnight. The membrane was fabricated by mixing 90 wt % zeolite power and 10 wt % polytetrafluoroethylene (PTFE, Sigma-Aldrich, 60 wt % in water). The mixture was ground evenly in a mortar and pressed into a freestanding thin film about 100 μm thin. After being cut into 14 mm disks, the zeolite membrane was dried under a vacuum at 120°C overnight. The dried membrane was stored in an Ar-filled glovebox before electrochemical measurements.

Preparation of Mo_3S_4 and PANI cathodes

Mo_3S_4 was synthesized by acid leaching from $Cu_2Mo_6S_8$ based on a previously reported procedure.[53] PANI was purchased from Fisher Scientific without any further purification. The Mo_3S_4 cathode was prepared by mixing 70 wt % active material, 20 wt % conductive carbon (Super P), and 10 wt % polyvinylidene difluoride (PVDF) in N-methyl-2-pyrrolidone (NMP; anhydrous, 99.5%; Sigma-Aldrich). The slurry was cast on carbon-coated aluminum foil (thickness: 18 μm) by doctor blading, and the electrodes were punched into 10 mm discs after being dried at 60°C for 5 h. The discs were finally dried *in vacuo* at 120°C overnight before being transferred into an Ar-filled glovebox. For the PANI cathode, 50 wt % PANI was mixed with 40 wt % conductive carbon (KETJENBLACK) and 10 wt % polytetrafluoroethylene (Sigma-Aldrich, 60 wt % dispersion in H_2O) in isopropanol. The freestanding PANI composite film thus obtained was dried at 60°C overnight and punched into small disks that were pressed onto a Ti mesh (diameter: 11 mm). These were utilized as cathodes in the full cells. The mass loading of the active material was ~2.5 mg cm^{-2}.

Electrochemical measurements

Mg foils (thickness: 100 μm) were polished by sandpaper several times before being pouched as 9/16-in disks and using them in symmetric or full 2,032 coin cells. To evaluate the effectiveness of the zeolite membrane, zeolite/Mg anodes were prepared by tightly pressing the zeolite membrane onto a Mg disc with hands before cell assembly. Such a contact between zeolite membrane and Mg was held by the pressure from spring in the coin cell. All cells—using 100 μL of electrolyte incorporated in a glass fiber separator—were cycled using a VMP3 potentiostat/galvanostat station (Bio-Logic). CV of the Mo_3S_4 cathode was measured at a sweep rate of 0.1 mV s^{-1} in a voltage range of 0.2–2.4 V. The evolution of the charge transfer resistance of two-electrode Mg symmetric cells was measured by EIS with a voltage amplitude of 5 mV in the frequency range of 1 MHz to 100 mHz. The cell was rested for 10 min to reach equilibrium before each EIS measurement. The electrochemical anodic stability of the electrolyte was investigated by linear sweep voltammetry at a sweep rate of 0.1 mV s^{-1}, based on three-electrode cells with different metal substrates with or without a zeolite membrane as both working and counter electrodes and Mg as the reference electrode. All cells were cycled at 25°C.

Materials characterization

Fourier transform infrared spectroscopy (FTIR) was carried out on a Bruker Tensor 27 system equipped with attenuated total reflection. A Zeiss Ultra field emission SEM instrument was used to collect SEM images and energy-dispersive X-ray spectroscopy (EDX) data. XPS experiments were conducted on a Thermo Scientific K-alpha XPS instrument. CasaXPS software was used to conduct XPS data analysis, where the C 1s peak of 284.8 eV was used to calibrate the binding energies. Spectral fitting was based on Gaussian-Lorentzian functions and a Shirley-type background. Solid-state ^{25}Mg magic angle spinning (MAS) NMR experiments were performed at 11.7 T (500 MHz) on a Bruker Avance III spectrometer operating at a Larmor frequency of 30.64 MHz using a 3.2 mm MAS probe. The spectra were acquired at a spinning speed of 5 kHz using 3.2 mm rotors in direct observe (single pulse) mode and with recycle delays of 1 s. ^{25}Mg shifts were referenced to 5 M $MgCl_2$ (aq.) at 0 ppm. All electrodes disassembled from the cells were washed three times with DME and dried at room temperature *in vacuo* before each characterization.

SUPPLEMENTAL INFORMATION

Supplemental information can be found online at https://doi.org/10.1016/j.joule.2023.10.012.

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AUTHOR CONTRIBUTIONS

C.L. and L.F.N. designed this study. C.L. carried out the characterization and all the electrochemical measurements. A.S. synthesized the Mg salt. B.K. conducted solid-state ^{25}Mg MAS NMR experiments. Z.Y. performed the TGA experiments. C.L. and L.F.N. wrote the manuscript with contributions from other authors.

DECLARATION OF INTERESTS

The authors declare no competing interests.

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